

InnerCheck: A New Breakthrough for Early Detection of Mental Health

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Abstract—Mental health is a global concern, with a high prevalence of mental disorders worldwide, inadequate treatment access, and specific challenges in addressing adolescent mental health in Indonesia. Despite advancements in the modern world, there is still a need for improved public understanding of mental health, as negative stigmas and misconceptions persist, hindering individuals from seeking proper care and support for their mental disorders, not only in developing countries like Indonesia but also in more advanced nations like the United States. This condition makes early detection crucial in addressing mental health problems. This paper aims to address these issues by developing Telegram chatbot. The Telegram chatbot is designed to detect early signs of mental health problems for timely intervention and support. The development of the Chatbot is using Scrum Framework. After completing the development, we conduct black box testing as a validation scheme. The result obtained is the creation of an InnerCheck Telegram chatbot can detect mental health problems and run according to the expected results based on black box testing.

Keywords—mental health, early detection, artificial intelligence, telegram chatbot

I. INTRODUCTION

Mental health is becoming an increasingly urgent and complex issue throughout the world, especially with the fact that nearly one billion people worldwide suffer from mental health disorders [1]. However, according to data from the World Health Organization (WHO), only half of them receive proper treatment [2]. Anxiety, eating disorders, bipolar, and schizophrenia are some of the mental health issues of increasing concern and are receiving severe attention worldwide.

Meanwhile, in Indonesia, data from the American Psychological Association (APA) shows that poor mental health is an issue often faced by Indonesian youth, with around 1 in 20 teenagers (about 5.5%) diagnosed with a mental disorder. This means that there are about 2.45 million teenagers throughout Indonesia who are included in the group of people with mental disorders [3]. Findings from the Indonesian National Mental Health Survey (I-NAMHS) even indicate that mental health problems in Indonesian adolescents are increasingly concerning, where one in three adolescents experience a mental health problem in the last 12 months and one in twenty adolescents meet the criteria for a mental disorder.

Adolescent mental health in Indonesia is becoming an increasingly critical issue because various kinds of mental disorders affect it. Anxiety disorders are the most common mental health problem among adolescents aged 10-17, affecting around 3.7% of the adolescent population. In addition, there are also major depressive disorders (1.0%), conduct disorder (0.9%), post-traumatic stress disorder (PTSD) and attention deficit hyperactivity disorder (ADHD), each of which affects 0.5% of the adolescent population [4].

A. Society's Stigma for People with Mental Disorders

The world is getting more sophisticated and modern. However, public understanding of mental health still needs to be improved. This is because there is still a lot of negative stigma in society, which makes people with mental disorders prefer to close themselves by not having their illness checked by a doctor or psychiatrist. For example, if a relative or neighbour has a mental disorder, many people will look down on sufferers. The practice of locking up, choosing to remain silent, hiding, and isolating people with mental disorders is still common. Not a few people with mental disorders are left on the street, wandering around, and are considered ashamed by their families [5]. These stigmas keep patients away from the best service they should be able to get. Negative stigma for people with mental disorders does not only occur in developing countries like Indonesia.

In the United States, which incidentally is more advanced, there is still debate about the relationship between mental health and violence. There is an opinion that suggests that cases such as mass shootings are related to mental health problems. This shows a negative stigma in society and that people who suffer from mental disorders are more likely to riot in the community. The facts show that people who suffer from mental disorders are not more likely to commit violence than people without mental disorders [6].

B. Mental Health Services in Indonesia

When Indonesia was still under colonial rule, mental health services only focused on exceptional mental hospitals. After independence in 1949, Indonesia inherited four large mental hospitals and several emergency clinics. Along with the development of the health system in Indonesia in the 1970s to 1990s, the number of specialized mental hospitals increased and various treatment methods were introduced. In addition, traditional mental health services such as shamans or intelligent people are also a concern in developing mental health services in Indonesia.

Along with the increasingly modern developments, the problems of mental health services in Indonesia are also increasingly diverse. The lack of psychiatrists and psychologists is one of the challenges [7]. In addition, the lack of facilities that can be used by people with mental health disorders in Indonesia also needs to be addressed. According to Director General of Public Health Maria Endang Sumiwi, there are still four provinces that do not have mental hospitals, and only 40% of public hospitals have psychiatric service facilities. As a result, people who suffer from mental health disorders cannot be detected and receive appropriate treatment [8].

These problems make innovation in mental health services even more critical. The development of technology that continues to move forward can be the right solution for people with mental disorders, both in the process of diagnosis and treatment. In this case, therapist automation technology is one of the answers that can make it easier to detect mental disorders earlier and prevent mental health crises from occurring.

The importance of these automated service providers cannot be ignored, especially in addressing the large demand for services in the current mental health crisis. This is even more important considering that in 2045, Indonesia is expected to have a golden generation with superior competence, strong mentality, integrity and good morals. Therefore, maintaining adolescent mental health must be a priority, and therapist automation technology can help [9].

Based on the description above, this paper develops a Telegram chatbot for the early detection of mental health problems. This paper aims to provide an early detection solution at a time when many cases of mental health are not well addressed due to suboptimal early detection, whether due to stigma or other factors. This will contribute to managing mental health issues, resulting in positive impacts on overall human well-being.

II. PRELIMINARY CONCEPT

A. Development of Artificial Intelligence

Artificial Intelligence (AI) technology is software that uses mathematical algorithms to process input information and produce relevant output [10]. One of the increasingly popular AI applications is chatbots, which are computer programs that can carry out written conversations with humans or other chatbots. Chatbot itself is an intelligent system resulting from Natural Language Processing (NLP), which is a branch of AI. In mental health therapy, AI technology and chatbots can provide more personalized and affordable support for patients. Still, it is essential to pay attention to data security and consider the advantages and disadvantages of using technology in mental health care [11].

AI developments and technological advances have made it possible to develop more intelligent and effective Telegram chatbots. Telegram, as a messenger platform, has provided an API (Application Programming Interface) that makes it easy for developers to create and integrate chatbots with other applications. This opens up the possibility for developers to create more complex chatbots with better ability to respond and process information. By using AI technology and APIs from Telegram, developers can create chatbots that are increasingly sophisticated and well-integrated within

applications, thereby providing a better user experience and meeting various needs, including mental health services.

B. Mental Therapy Using Chatbot Technology

In the context of mental health therapy, technology can be used as an effective tool in assisting therapists and patients in overcoming various problems they face, such as mental health applications that provide easy access for patients to seek help anytime and anywhere. In addition, virtual therapy or teletherapy can also facilitate remote treatment and increase the effectiveness of therapy. Modern technologies such as AI and chatbots have also been used in mental health therapy to provide more personalized and affordable support. The Telegram chatbot can act as a tool in therapy sessions by asking various questions every day to see if the user is experiencing symptoms of a mental disorder.

There is already an example of chatbot technology used for mental health therapy, called Woebot. Woebot is a chatbot equipped with artificial intelligence technology designed to help users deal with stress and anxiety. Woebot can build relationships with users and provide emotional support in a similar way to therapy sessions with a human therapist.

Even so, Woebot itself still has some drawbacks. Among other things, users must download the application to use its features. In addition, Woebot also requires its users to create an account first, so it tends to be less secure for user privacy. Another drawback, for now, Woebot is only available on the App Store and Google Play, so Woebot can only be downloaded on personal devices [12].

III. RESEARCH METHOD

In carrying out the Telegram chatbot development process, Scrum method is used. Scrum is one of the methods of device development. The fastest and most flexible software. The planning and development phase, which includes implementation, testing, and evaluation, can be done simultaneously and quickly [18].

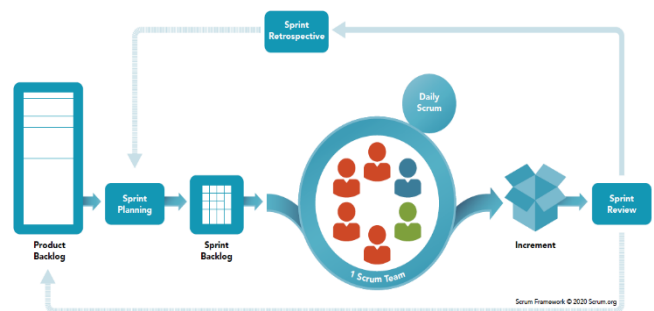


Fig. 1. Scrum Framework [16]

The Telegram chatbot uses the Scrum method in two phases: planning and development. There are four stages of the two phases: sprint planning, sprint development (daily scrum), sprint review, and sprint retrospective. Each of the two phases is explained as follows.

- 1) Sprint Planning is the first step in planning the project implementation. It starts with the product owner talking about the product backlog, then the sprint backlog, or a

list of assignments that correspond to the product backlog that has been delivered. The product manager assigns those tasks to each team member.

- 2) Sprint Development involves implementing the backlog list created in the planning stage. To synchronize project progress with the rest of the team, daily meetings and weekly meetings are held.
- 3) Sprint Review involves evaluating the results of the implementation, which is done both weekly and within a sprint. It also presents the results of feature testing on the product specifications implemented in the development phase.
- 4) Sprint Retrospective is a stage where each team member performs tasks to assess their performance during the development stage. In addition, the entire team talks about the problems encountered while working on the task, as well as the solutions to those problems. This can be used as learning and improve the efficiency of the task in the next sprint.

Furthermore, the black box testing method will be used after the development process is complete [17]. Black box testing will be conducted on the Telegram chatbot system and focus on the functional needs of the application. Black box testing will check whether the input given matches the desired output.

IV. RESULT AND DISCUSSION

A. New Breakthroughs for Mental Health

The limitations of Woebot make the Telegram chatbot a promising alternative for mental health detection. The main reason why Telegram can be a breakthrough is that Telegram can be accessed through all platforms, from personal devices, laptops, computers, and websites. With the availability of bots across multiple devices, Telegram chatbots can reach more people.

The next reason is that the Telegram chatbot uses artificial intelligence, making it an easily accessible and anonymous resource for mental health support. Not only that, Telegram chatbots can provide a safer platform for individuals for mental health support because users do not need to create separate accounts to use them. Users can utilize their existing Telegram accounts to access the bot. This can help address various mental health-related problems, such as accessibility and discomfort when discussing personal issues with strangers.

Telegram chatbot has various essential functions in managing mental health. Here are the functions of the Telegram chatbot in mental health:

- The Telegram chatbot can provide support through AI-powered conversation therapy. This chatbot can help users understand their mental health issues safely and effectively. According to a study, conversation therapy through a chatbot can help reduce symptoms of depression and anxiety in users [13].
- The Telegram chatbot can provide information about symptoms of mental disorders. The chatbot can help users understand the symptoms they are experiencing. According to a study, chatbots can assist users in acquiring knowledge and understanding of mental disorders [14].

- The Telegram chatbot can provide advice and support to manage stress and anxiety. According to a study, a chatbot equipped with relaxation techniques can help reduce stress levels in users.
- The Telegram chatbot can provide resources and references to obtain professional help. The chatbot can assist users in finding resources and references for accessing professional assistance, such as psychologists or therapists. According to a study, chatbots can help users find information about mental health services and facilitate their search for help [15].

B. Telegram Chatbots Development

The Telegram chatbot we designed is named InnerCheck. "InnerCheck" is interpreted as a bot that assists in examining or evaluating a person's inner condition, which refers to their mental or emotional state. "Inner" refers to aspects related to oneself, such as feelings, thoughts, and emotions. "Check" means examination or evaluation. So, overall, "InnerCheck" can be understood as a bot that helps in examining or evaluating someone's inner or mental condition.

The development process of the chatbot starts with the design and coding of algorithms to create a bot that can provide automated responses to user inputs. InnerCheck is built using simple programming in the Python language. The brief algorithm for InnerCheck is as follows:

- 1) Users access the bot through Telegram.
- 2) Users choose the bot menu they want to use, with three menu options available: /start, /question, and /help.
- 3) If users choose /start, the bot will display an opening message that includes instructions on how to use the bot.
- 4) If users choose /question, the bot will present a list of questions that users must answer with a yes or no option.
- 5) If users choose /help, the bot will provide a brief explanation of the available menus in InnerCheck.

Next, once the user has answered all the questions provided by the InnerCheck bot, the bot will calculate the results and display an initial assessment of the user's mental health. If the bot detects a mental disorder in the user, it will guide the user to consult with the nearest psychiatrist. If the bot does not detect any mental disorder, it will display a message to always prioritize mental health. The details of the process can be seen in Fig. 1 below.

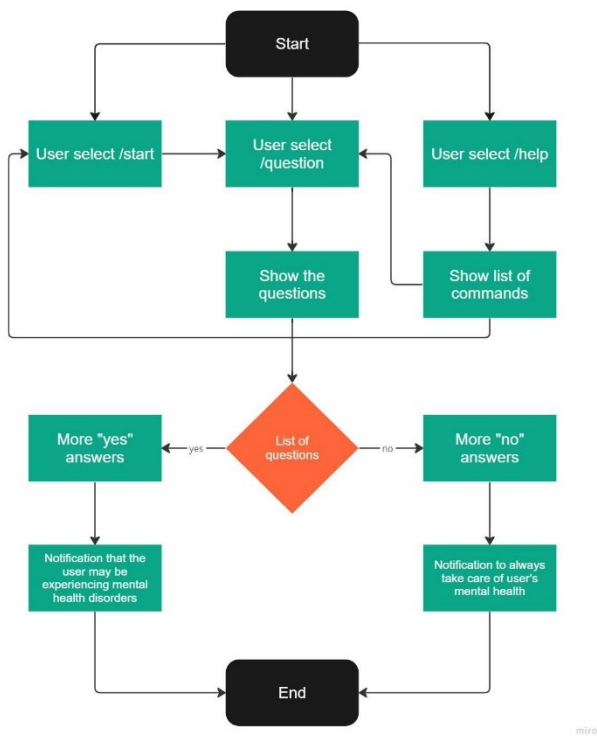


Fig. 2. InnerCheck chatbot flowchart

Some of the chatbot interface displays can be seen as follows:

- Chatbot initial view: This is the initial display of the InnerCheck chatbot, where users will be greeted with a brief explanation of the features that the bot can perform. Afterwards, users will be prompted to press the start button to initiate the bot.

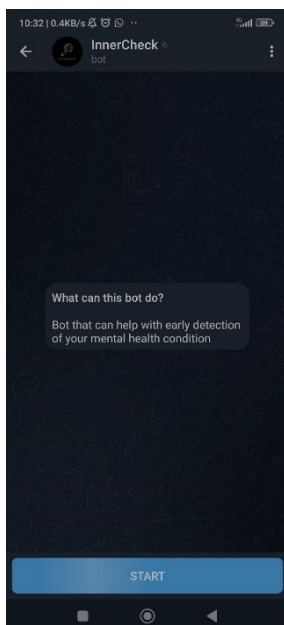


Fig. 3. InnerCheck chatbot initial view

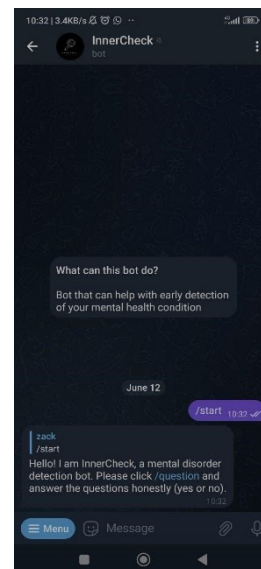


Fig. 4. Start view

- Chatbot Display If the User Is Not Indicated to Have a Mental Disorder: This is the display of the InnerCheck chatbot when the user is not detected to have a mental disorder.

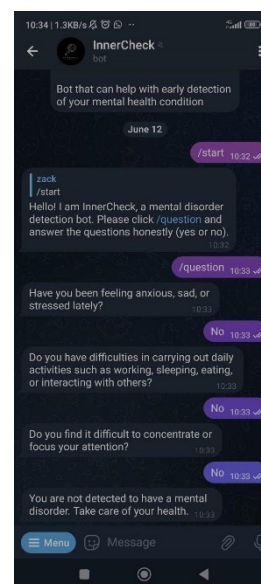


Fig. 5. InnerCheck chatbot display if the user is not indicated to have a mental disorder

- Chatbot Display If a User Is Indicated to Have a Mental Disorder: This is the display of the InnerCheck chatbot when the user is detected to have a mental disorder.

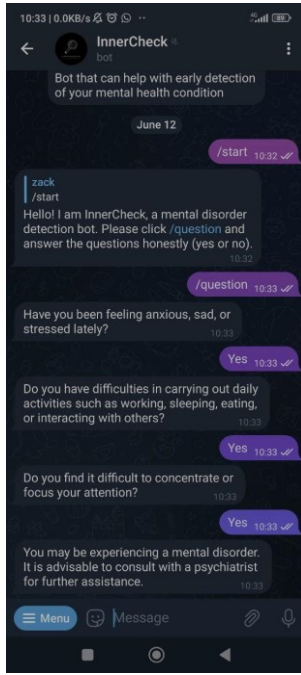


Fig. 6. InnerCheck chatbot display if the user is indicated to have a mental disorder

C. Blackbox Testing

Testing begins with the creation of a test case. The test case contains the steps performed. The list of test cases and expected results is shown in Table I.

TABLE I. BLACK BOX TESTING PLANNING

ID Test Case	Test Case	Expected Result
001	Access username	System shows chatbot initial display
002	Access menu	System shows command
003-01	Start bot	System displays notification
003-02	Press 'yes' button	System does not display any notification
003-03	Press 'no' button	System does not display any notification
004-01	Input command/question	System displays questions
004-02	Press 'yes' button after input command/question	System passes to next question
004-03	Press 'no' button after input command/question	System passes to next question
004-04	Finish answering all questions	System shows test result
005	Input command/help	System shows help display

After knowing the design of black box testing, the next step is to conduct tests and record various test results. The list of test results and their conclusions can be seen in Table II the following.

TABLE II. BLACK BOX TESTING RESULT

ID Test Case	Test Result	Conclusion
001	System shows the chatbot initial display	Match
002	System shows command	Match
003-01	System displays notification	Match
003-02	System does not display any notification	Match
003-03	System does not display any notification	Match
004-01	System displays questions	Match
004-02	System passes to next question	Match
004-03	System passes to next question	Match
004-04	System shows test result	Match
005	System shows help display	Match

Based on the tests that have been done, the total number of data obtained is as many as ten tests on five functions, namely viewing chatbots, chatbot details, entering commands in the form of start, question, help, and answer all questions to see the results of early detection of users' mental health. The test results show that all functions are following with the expected results.

D. Advantages of Mental Health Therapy with Chatbot Telegram

Through the integrated chatbot on Telegram, there are several advantages that users can obtain, including:

- **Speed and convenience**
Telegram chatbot is very fast in responding to user requests, allowing users to receive assistance or support anytime and anywhere without waiting a long time. Additionally, Telegram has a simple and user-friendly interface, making users feel comfortable and at ease when interacting with the chatbot.
- **Confidentiality and privacy guaranteed**
Telegram consistently ensures the privacy and confidentiality of user information. Users can interact with the Telegram chatbot anonymously and do not need to worry about the privacy and security of their personal information.
- **Easy to access and flexible**
Telegram can be easily accessed through applications available on various platforms, such as desktop, mobile, and web. This allows users to engage in mental health therapy flexibly, without limitations of time and place.
- **Integration and development capabilities**
Telegram has an API that enables efficient development and management of chatbots. Additionally, Telegram has the ability to integrate with third-party applications, allowing users to access various additional services that can support their mental health therapy.

V. CONCLUSION AND FUTURE WORK

This research successfully developed a Telegram chatbot used for the early detection of health issues that can arise in anyone. InnerCheck is built using the Scrum Model and has gone through black box testing with good test results. With InnerCheck, it is hoped that mental health problems can be detected as early as possible so that they get the proper treatment. The development of the chatbot named InnerCheck provides several advantages such as speed, convenience, confidentiality, privacy guaranteed, accessibility, flexibility, and integration and development capabilities. Regarding the development of the chatbot, this paper explains various aspects, including the algorithm, flowchart, and user interface of the chatbot during its usage. The expected future work of this paper is the development of the Telegram chatbot to integrate with various other services, such as location maps of mental health facilities like psychiatric hospitals or psychologists. Additionally, the existing features can be expanded or enhanced according to future needs.

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